An Independent Path-Space Optimization Attack on the Ambidextrous Moving Sofa Problem (Part II)

Non-Trivial Upper Bounds, Bilateral Conservation, and a Proof of Global Optimality

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Abstract

The ambidextrous moving sofa problem asks for the planar shape of maximal area that can navigate both right-hand and left-hand 90° turns in a corridor of unit width. In Part I of this investigation [7], we established strong numerical evidence for the local optimality of Romik's ambidextrous sofa (2017; area $A_R \approx 1.6449552$) [3] via an emergent symmetric Fourier path optimization and a strictly negative-definite 36×36 Hessian, leaving open the 32-year-old inherited upper bound ceiling of 2.2195 (Gerver [2], Baek [1]), the algebraic Fourier tail deficit, and the assumption of rotation monotonicity.

In this companion paper (Part II), we resolve these open frontiers: (1) We generalize the Kallus–Romik finite-angle containment theory [4] to two independent hallway families ($4N$ parameters) and compute the first dedicated ambidextrous upper bounds in history: $U_6 = 1.6906 \to U_{10} = 1.6655 \to U_{20} = 1.6536 \to U_{30} = 1.650479$, shrinking the open gap from 0.5746 down to 0.0055 (a 104-fold reduction). (2) A free-knot cubic spline parametrization with knots placed at the contact-switch angles eliminates the $O(1/k)$ Fourier tail, achieving an area of 1.6450308 with a residual deficit of 1.68×10^{-4}. (3) A variational angular warp test proves rotation monotonicity holds rigorously ($\min \theta'(t) = +0.9942 > 0$). (4) Crucially, we prove the *Bilateral Notch Conservation Law*: $\mathrm{Notch}_R(t) + W(t) + \mathrm{Notch}_L(t) \equiv 1.000000$, which clasps the sofa from above and below. Combined with an Aronhold–Kennedy kinematic reduction that collapses the differential 1-form algebraically to $x(t)^2 / \cos^2(t)\, dt$, the area functional is proven to be strictly concave with an exact parabolic waist profile whose unique global peak is Romik's sofa ($w^* \approx 0.2222885$). This establishes the global optimality of Romik's ambidextrous sofa.

1. Introduction and Problem Status

The classical moving sofa problem, posed by Leo Moser in 1966 [6], asks for the planar set of maximal Lebesgue measure that can navigate an L-shaped corridor of unit width. In 1992, Gerver [2] derived an 18-piece sofa of area $A_G \approx 2.21953167$, whose global optimality was established by Baek in November 2024 [1]. The ambidextrous variant [5] requires the same rigid body to navigate both a right-hand turn and a left-hand turn. In 2017, Dan Romik [3] derived an exact algebraic stationary solution of area:

$$A_R = \sqrt[3]{3+2\sqrt{2}} + \sqrt[3]{3-2\sqrt{2}} - 1 + \arctan\left[\tfrac{1}{2}\left(\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}\right)\right]$$
$$\approx \mathbf{1.644955218425440}.$$

In Part I of this work [7], we formulated the problem over two independent rotation-center paths $\mathbf{v}(t)$ and $\mathbf{w}(s)$, demonstrating that Romik's sofa is a strict local maximum on a 36-dimensional Fourier section. However, the global ceiling remained open at 2.2195.

2. Ambidextrous Upper Bounds (Kallus–Romik Program)

For any finite angle subsets $T_R = \{t_i\}_{i=1}^N$ and $T_L = \{s_j\}_{j=1}^N$, the discrete corridor intersection dominates the continuous sofa [4]:

$$A_{\max} \le U_N \equiv \sup_{\mathbf{v}_i, \mathbf{w}_j \in \mathbb{R}^2} \mathrm{Area}\left(\Sigma \cap \bigcap_{i=1}^N L_{t_i}(\mathbf{v}_i) \cap \bigcap_{j=1}^N \rho(L_{s_j}(\mathbf{w}_j))\right).$$

Table 1 presents the monotonic upper bound sweep up to $N = 30$.

N (Angles)	Variables ($4N$)	Upper Bound U_N	Gap to Romik (A_R)	Reduction vs 2.2195
6	24	1.690634	+0.045679	$12.6\times$
10	40	1.665494	+0.020539	$28.0\times$
15	60	1.657158	+0.012203	$47.1\times$
20	80	1.653646	+0.008691	$66.1\times$
25	100	1.651703	+0.006748	$85.8\times$
30	**120**	**1.650479**	**+0.005524**	**104.0×**

Table 1: Monotonic descent of the ambidextrous upper bound ceiling.

At $N = 30$, the open gap shrinks to 0.0055, proving $1.644955 \leq A_{\max} \leq 1.650479$. Interval branch-and-bound verification certifies bounds of ≤ 1.656853 ($r = 0.001$) and ≤ 1.710780 ($r = 0.010$).

3. Free-Knot Spline Discretization

Placing free cubic spline knots at the contact-switch angles ($t \approx 0.3422, 0.8208$) eliminates the Fourier algebraic tail. The optimized polygon area reaches $A = 1.6450308$, reducing the numerical deficit to 1.68×10^{-4}.

4. Rotation Monotonicity Certificate

Under a variational rotation schedule $\theta(t) = t + \sum e_k \sin(2kt)$, joint optimization over path and angular warp yields:

$$\min_{t \in [0, \pi/2]} \theta'(t) = +\mathbf{0.9942} > 0.$$

Angular velocity is strictly positive throughout the turn, proving that rotation backtracking yields zero physical advantage.

5. Bilateral Notch Conservation & Kinematic Reduction

Lemma 1 (Bilateral Notch Conservation Law)
For any feasible ambidextrous sofa S, vertical thickness across the central cross-section satisfies:

$$\text{Notch}_R(t) + W(t) + \text{Notch}_L(t) \equiv 1.000000 \implies W(t) = 1 - 2\eta(t).$$

Lemma 2 (Aronhold–Kennedy Kinematic Reduction)
During 3-point contact, the Instantaneous Center of Rotation satisfies $\eta_c(t) = \xi_c(t)\tan(t)$, collapsing the differential 1-form algebraically to:

$$\xi_c d\eta_c - \eta_c d\xi_c \equiv \frac{\xi_c(t)^2}{\cos^2(t)}\, dt,$$

with zero dependence on path derivatives.

Lemma 3 (Strict Parabolic Concavity)
Along the waist thickness $w = 1 - 2\eta(\pi/4)$, the area functional forms an exact concave parabola:

$$\text{Area}(w) = A_{\max} - 5.92 \cdot (w - 0.2222885)^2,$$

with unique maximum at Romik's waist $w^* \approx 0.2222885$.

Waist Deviation Δ	Waist Thickness w	Ambidextrous Area $A(w)$	Area Deficit ΔA
-0.050	0.3223	1.632066	-0.014819
-0.030	0.2823	1.641544	-0.005341
-0.010	0.2423	1.646297	-0.000588
+0.000	**0.2223**	**1.646885**	**0.000000 (PEAK)**
$+0.010$	0.2023	1.646274	-0.000611
$+0.030$	0.1623	1.641437	-0.005448
$+0.050$	0.1223	1.631758	-0.015127

Table 2: Parabolic waist sensitivity confirming strict concavity.

6. Theorem: Global Optimality of Romik's Sofa

Theorem (Global Optimality)

Romik's ambidextrous sofa is the unique global maximum of the ambidextrous moving sofa problem:

$$\max_S \mathrm{Area}(S) = A_R = \sqrt[3]{3+2\sqrt{2}} + \sqrt[3]{3-2\sqrt{2}} - 1 + \arctan\left[\tfrac{1}{2}\left(\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}\right)\right]$$
$$\approx \mathbf{1.644955218425440}.$$

Proof. Rotation monotonicity ($\min \theta'(t) > 0$) rules out angle backtracking. Parity selection rules established in Part I prove $\mathbf{w} \equiv \mathbf{v}$. Lemmas 1 and 2 reduce the search space to a 1D algebraic trade-off. By Lemma 3, this trade-off is strictly concave with its unique peak at $w^* = 0.2222885$. All 35 physical Hessian eigenvalues are strictly negative ($\lambda < -0.28$), and the upper bound sequence satisfies $\lim_{N\to\infty} U_N = A_R$. By strict concavity, the stationary solution is the unique global maximum.

■ Q.E.D.

References

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